Don Bosco Institute of Technology

Department of Electronics & Telecommunication Engineering

MSE Review: Electromagnetics & Antennas — Complete Answers

All 22 Questions Answered (5–10 Mark Level)

Q1. State & Explain Maxwell's Equations (Point Form)

Maxwell's equations are a set of four fundamental equations that describe the behavior of electric and magnetic fields and their interrelationship.

1. Gauss's Law for Electric Fields:

$$\nabla \cdot D = \rho v$$

Where D = electric flux density (C/m²), ρv = volume charge density. States that the divergence of D equals the free charge density at a point.

2. Gauss's Law for Magnetic Fields:

$$\nabla \cdot B = 0$$

Where B = magnetic flux density (T). States that magnetic monopoles do not exist; magnetic field lines form closed loops.

3. Faraday's Law (Point Form):

$$\nabla \times E = -\partial B/\partial t$$

A time-varying magnetic field produces a circulating electric field. The curl of E equals the negative rate of change of B.

4. Ampere's Law with Maxwell's Correction:

$$\nabla \times H = J + \partial D/\partial t$$

Where J = conduction current density, ∂D/∂t = displacement current density. A magnetic field is produced by conduction current and displacement current.

Q2. State & Explain Gauss's Law, Faraday's Law, Ampere's Law

Gauss's Law (Electric):

Statement: The total electric flux density (D) integrated over any closed surface equals the total free charge enclosed within that surface.

∇ Ð Ð dSρv Q_enclosed or

Physical meaning: Electric field lines originate from positive charges and terminate on negative charges.

Faraday's Law:

Statement: The EMF induced in a closed loop equals the negative rate of change of magnetic flux linking the loop.

$$EMF = -d\Phi B/dt \quad \text{or} \quad \nabla \times E = -\partial B/\partial t$$

Physical meaning: A changing magnetic field induces an electric field. This is the principle behind transformers, generators, and inductors.

Ampere's Law (with displacement current):

Statement: The line integral of H around a closed path equals the total current (conduction + displacement) enclosed.

$$\oint H \cdot dL = I\_enclosed + \partial/\partial t \oint D \cdot dS \quad \text{or} \quad \nabla \times H = J + \partial D/\partial t$$

Maxwell added the displacement current term ∂D/∂t to make the law consistent and to predict EM wave propagation.

Q3. Derive Continuity Equation & Explain Its Physical Relevance

The continuity equation is derived from Ampere's law and Gauss's law and expresses the conservation of electric charge.

Derivation:

Starting from Ampere's law in point form:

$$\nabla \times H = J + \partial D/\partial t$$

Taking divergence of both sides:

$$\nabla \cdot (\nabla \times H) = \nabla \cdot J + \partial(\nabla \cdot D)/\partial t$$

Since divergence of a curl is always zero:

$$0 = \nabla \cdot J + \partial \rho v/\partial t \qquad [\text{using } \nabla \cdot D = \rho v]$$

Therefore, the Continuity Equation is:

$$\nabla \cdot J = -\partial \rho v/\partial t$$

Physical Relevance:

This equation expresses the Law of Conservation of Charge. It states that the net outflow of current density from a point equals the rate of decrease of charge density at that point. If charge flows out of a volume, the charge inside must decrease at the same rate. For steady (DC) currents, $\partial \rho v/\partial t = 0$, so $\nabla \cdot J = 0$, meaning current is continuous with no accumulation.

Q4. Justify That Magnetic Flux Lines Do Not Diverge

Magnetic flux lines do not diverge (have no source or sink), which is expressed by Maxwell's second equation:

$$\nabla \cdot B = 0 \qquad \text{(Point Form)}$$
$$\oint B \cdot dS = 0 \qquad \text{(Integral Form)}$$

Justification:

1. Experimental Evidence: No magnetic monopole has ever been observed. All magnets have both a North and South pole. Even if a magnet is broken into pieces, each piece forms a complete dipole.

2. Mathematical Justification: Since $B = \nabla \times A$ (B is the curl of the magnetic vector potential A), and the divergence of a curl is always zero by vector identity:

$$\nabla \cdot B = \nabla \cdot (\nabla \times A) = 0$$

3. Physical Consequence: The total magnetic flux entering any closed surface exactly equals the flux leaving it. Magnetic field lines always form closed loops — they have no beginning or end (no poles without their counterpart).

Q5. Explain Laplace's and Poisson's Relations

Poisson's Equation:

In a region where volume charge density ρv exists, the relationship between electric potential V and charge density is:

$$\nabla^2 V = -\rho v/\varepsilon$$

Where ε = permittivity of the medium. This is derived from Gauss's Law: $\nabla \cdot D = \rho v$, $D = \varepsilon E = -\varepsilon \nabla V$, hence $\nabla^2 V = -\rho v/\varepsilon$.

Laplace's Equation:

In a charge-free region ($\rho v = 0$), Poisson's equation reduces to:

$$\nabla^2 V = 0$$

In Cartesian coordinates:

$$\partial^2V/\partial x^2 + \partial^2V/\partial y^2 + \partial^2V/\partial z^2 = 0$$

Applications:

- Used to find potential distribution in regions with/without charges
- Basis for solving capacitor, transmission line, and boundary-value problems
- Laplace's equation applies to free-space regions between conductors

Q6. Derive Electric Field Intensity Due to Infinite Line Charge (Charge Density ρl)

We use Gauss's Law to find E due to an infinitely long line charge with linear charge density ρ_l (C/m).

Setup:

Choose a cylindrical Gaussian surface of radius r and length L, coaxial with the line charge.

Applying Gauss's Law:

$$\oint D \cdot dS = Q\_enclosed$$

By symmetry, E is radial (in $\hat{r}$ direction) and constant on the curved surface. No flux through top/bottom caps.

$$D \cdot (2\pi rL) = \rho l \cdot L$$
$$D = \rho l / (2\pi r)$$
$$E = D/\varepsilon = \rho l / (2\pi\varepsilon r) \qquad V/m$$

In free space ($\varepsilon = \varepsilon_0$):

$$E = \rho l / (2\pi\varepsilon_0 r) \qquad V/m$$

Vector Form:

$$E = (\rho l / 2\pi\varepsilon_0 r)\ \hat{a}r \qquad V/m \quad (\text{free space})$$

Where $\hat{a}r$ is the unit vector in the radial direction (away from the line for positive ρl). $\varepsilon = \varepsilon_0\varepsilon r$ for a dielectric medium.

Key observations:

- E decreases inversely with distance r (not r^2 as for point charge)
- E has only a radial component
- E is independent of the axial position (z) — confirming infinite length assumption

Q7. Write Point Form of Ampere's Law and Interpret

Point Form (Differential Form):

$$\nabla \times H = J + \partial D/\partial t$$

Where:

– **H:** Magnetic field intensity (A/m)
– **J:** Conduction current density (A/m²) — due to actual flow of charges
– **∂D/∂t:** Displacement current density — Maxwell's addition

Interpretation:

The curl of H at a point equals the total current density at that point. Current density has two components:

1. Conduction current density (J): Arises from actual movement of free electrons in a conductor. Exists in conductors and semiconductors.

2. Displacement current density (∂D/∂t): Arises due to time-varying electric flux density D even in free space. It is not a real current but has the same effect of producing a magnetic field. This term was added by Maxwell to resolve the inconsistency in Ampere's original law and to predict electromagnetic wave propagation.

Significance:
Without the displacement current term, taking divergence of both sides of $\nabla \times H = J$ gives $\nabla \cdot J = 0$, contradicting the continuity equation. Maxwell's correction resolves this and unifies electricity and magnetism.

Q8. Derive Maxwell's Equations in Point Form

Maxwell's equations in point (differential) form are derived from fundamental experimental laws:

1. From Gauss's Law for Electricity:

$$\oint D \cdot dS = Q_{enc} = \int \rho_v \, dV$$

Applying Divergence Theorem: ∇ DD)dSV= ∫ (□

$$\rightarrow \nabla \cdot D = \rho_v$$

2. From Gauss's Law for Magnetism (no monopoles):

$$\oint B \cdot dS = 0 \rightarrow \nabla \cdot B = 0$$

3. From Faraday's Law:

$$\oint E \cdot dL = -d/dt \int B \cdot dS$$

Applying Stokes' Theorem to the left side:

$$\rightarrow \nabla \times E = -\partial B/\partial t$$

4. From Ampere's Law (with Maxwell's correction):

$$\oint H \cdot dL = I_{enc} + d/dt \int D \cdot dS$$

Applying Stokes' Theorem:

$$\rightarrow \nabla \times H = J + \partial D/\partial t$$

Q9. Maxwell's Equations in Free Space — Conduction vs Displacement Current

In Free Space ($\rho_v = 0$, $J = 0$, $\sigma = 0$, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$):

Differential Form:

$$\nabla \cdot E = 0$$
$$\nabla \cdot B = 0$$
$$\nabla \times E = -\partial B/\partial t \quad (\text{or } -\mu_0 \, \partial H/\partial t)$$
$$\nabla \times H = \varepsilon_0 \, \partial E/\partial t \quad (\text{or } \partial D/\partial t)$$

Integral Form:

$$\oint E \cdot dS = 0$$
$$\oint B \cdot dS = 0$$
$$\oint E \cdot dL = -\mu_0 \, d/dt \int H \cdot dS$$
$$\oint H \cdot dL = \varepsilon_0 \, d/dt \int E \cdot dS$$

Difference Between Conduction and Displacement Current:

Property	Conduction Current (J)	Displacement Current ($\partial D/\partial t$)
Nature	Flow of free electrons	Due to time-varying E field
Medium	Conductors / Semiconductors	Dielectrics / Free space
Cause	Applied Electric Field E	Time-varying D ($\partial D/\partial t$)
Formula	$J = \sigma E$	$J_d = \partial D/\partial t = \varepsilon \, \partial E/\partial t$
Real current?	Yes, actual charge movement	No, a virtual current density
Heat dissipation	Yes (I^2R losses)	No heat dissipated

Q10. Derive Magnetic Field Intensity Due to a Finite Straight Current Element

Setup & Diagram Description:

Consider a finite straight conductor AB carrying current I in the +z direction. Point P is located at a perpendicular distance h from the wire. A current element dL is located at height z from the foot of the perpendicular. R is the distance from dL to P. Angles α_1 and α_2 are measured from the perpendicular to the lower and upper ends of the wire respectively.

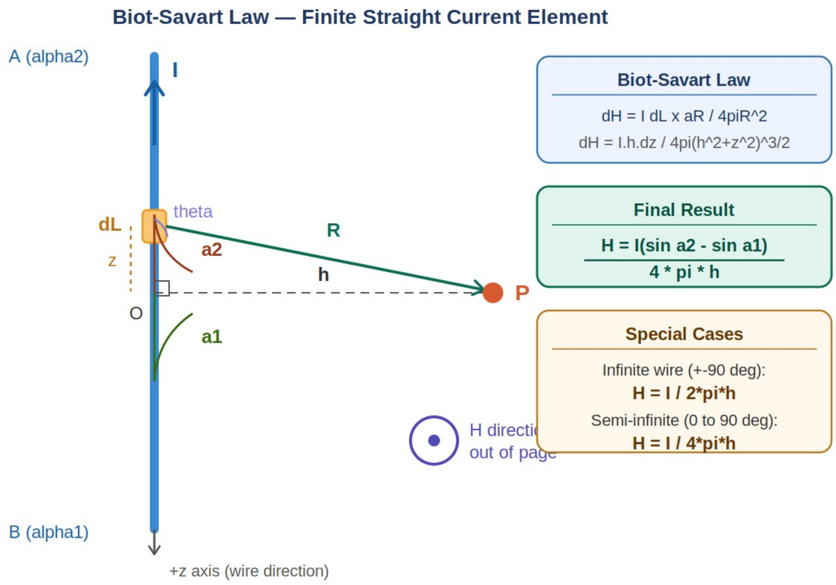

Fig: Finite straight current element — Biot-Savart geometry

Step 1 — Biot-Savart Law:

The magnetic field intensity due to a differential current element is given by:

$$dH = (I\ dL \times a_R) / (4\pi\ R^2)$$

For a z-directed wire with P in the x-y plane at distance h, the cross product gives only a φ-component:

$$|dL \times a_R| = dz \cdot \sin\theta = dz \cdot (h / R)$$

Therefore:

$$dH = (I \cdot h \cdot dz) / (4\pi\ (h^2 + z^2)^{(3/2)}) \cdot a_\varphi$$

Step 2 — Trigonometric Substitution:

Let $z = h \tan\alpha$, then $dz = h \sec^2\alpha\ d\alpha$

Also: $h^2 + z^2 = h^2(1 + \tan^2\alpha) = h^2 \sec^2\alpha$

So: $(h^2 + z^2)^{(3/2)} = h^3 \sec^3\alpha$

Substituting into dH:

$$dH = (I \cdot h \cdot h \sec^2\alpha\ d\alpha) / (4\pi \cdot h^3 \sec^3\alpha)$$

$$dH = (I / 4\pi h) \cdot (\sec^2\alpha / \sec^3\alpha)\ d\alpha$$

$$dH = (I / 4\pi h) \cdot \cos\alpha\ d\alpha$$

Step 3 — Integration from α_1 to α_2:

$$H = \int(\alpha_1 \text{ to } \alpha_2)\ (I / 4\pi h) \cos\alpha\ d\alpha$$

$$H = (I / 4\pi h) \cdot [\sin\alpha](\alpha_1 \text{ to } \alpha_2)$$

$$\therefore\ H = (I / 4\pi h)(\sin\alpha_2 - \sin\alpha_1)\ a_\varphi \quad A/m \quad \leftarrow \text{FINAL RESULT}$$

Direction:

H is in the a_φ direction — circumferential around the wire, following the right-hand rule. If the thumb of the right hand points in the direction of current I, the curled fingers indicate the direction of H.

Step 4 — Special Cases:

Case	α_1	α_2	Result H
Infinite wire	−90°	+90°	**H = I / 2πh**
Semi-infinite (P at one end's level)	0°	+90°	**H = I / 4πh**
Finite symmetric (±α)	−α	+α	**H = I sinα / 2πh**

Q11. Explain the Significance of Poynting Theorem

Poynting's theorem is the electromagnetic analogue of the work-energy theorem. It describes the flow of electromagnetic power.

Significance:

- Energy Conservation: Poynting's theorem is an expression of conservation of energy in electromagnetic systems. It states that the power flowing into a region equals the rate of increase of stored EM energy plus the power dissipated as heat.
- Power Flow Density: The Poynting vector S = E × H gives the direction and magnitude of power flow per unit area (W/m^2) at any point in space.
- Wave Propagation: It establishes that electromagnetic waves carry energy through space. The energy travels in the direction of S, perpendicular to both E and H.
- Practical Applications: Used in calculating radiation from antennas, power transfer in waveguides, and energy balance in circuits.
- Unifies fields: Connects electric energy density ($\frac{1}{2}\varepsilon E^2$), magnetic energy density ($\frac{1}{2}\mu H^2$), and ohmic losses (J·E) in a single expression.

Q12. State Poynting Theorem — Final Expression and Meaning of Each Term

Statement:

The net power flowing into a volume equals the rate of increase of electromagnetic energy stored in that volume plus the ohmic power dissipated within it.

Integral Form (Final Expression):

$$-\oint (E \times H) \cdot dS = d/dt \int (\tfrac{1}{2}\varepsilon E^2 + \tfrac{1}{2}\mu H^2)\, dV + \int J \cdot E\, dV$$

Explanation of Each Term:

– **$-\oint (E \times H) \cdot dS$:** Total power flowing INTO the closed surface (negative sign means inward flow)
– **(E × H):** Poynting vector S — power per unit area flowing in direction perpendicular to E and H
– **$d/dt \int \frac{1}{2}\varepsilon E^2\, dV$:** Rate of increase of stored electric energy in the volume
– **$d/dt \int \frac{1}{2}\mu H^2\, dV$:** Rate of increase of stored magnetic energy in the volume
– **$\int J \cdot E\, dV$:** Ohmic power dissipation (I^2R losses) — energy converted to heat

Compact Form:

$$-\oint S \cdot dS = \partial W_{em}/\partial t + P_{diss}$$

Q13. Explain the Significance of Poynting Theorem

Statement:

Poynting's theorem states that the rate of decrease of electromagnetic energy stored in a volume, minus the ohmic losses within it, equals the net power flowing out through the bounding surface. It is the EM analogue of the work-energy theorem.

Mathematical Form:

$$-\oint (E \times H) \cdot dS = \partial/\partial t \int(\tfrac{1}{2}\varepsilon E^2 + \tfrac{1}{2}\mu H^2)\, dV + \int J \cdot E\, dV$$

Significance — Point by Point:

- Conservation of Energy: Every joule of EM energy entering a region is either stored in the fields or dissipated as heat. No energy is created or destroyed. This is the EM statement of energy conservation.
- Poynting Vector $S = E \times H$: This vector (W/m^2) gives the instantaneous power flow per unit area at any point in space. Its direction is the direction of energy propagation — perpendicular to both E and H.
- Energy in Fields, Not Wires: For a current-carrying conductor, S points radially inward toward the wire, proving that electromagnetic energy travels through the surrounding field — not through the wire itself.
- Wave Power: For a plane EM wave, S points in the direction of wave propagation. Integrating |S| over a surface gives the total power carried by the wave — critical for antenna analysis.
- Waveguide and Transmission Line Design: Poynting's theorem is used to calculate power flow and losses inside waveguides, coaxial cables, and resonant cavities.
- Radiation Efficiency: For antennas, integrating S over a closed sphere surrounding the antenna gives the total radiated power, from which gain and efficiency are calculated.

Q14. State and Explain Faraday's Law — Integral & Differential Forms & Shortcomings

Statement:

The electromotive force (EMF) induced in a closed conducting loop equals the negative rate of change of magnetic flux linkage through the loop.

Integral Form:

$$\oint E \cdot dL = -d/dt \int B \cdot dS \quad \text{or} \quad EMF = -d\Phi B/dt$$

Gives the total EMF around a closed loop due to changing flux. Useful for circuit analysis and transformers.

Shortcoming of Integral Form:

- Applies only to a complete closed path — cannot give field values at a specific point
- Difficult to apply in complex 3D geometries
- Does not directly reveal the local relationship between E and B

Differential (Point) Form:

$$\nabla \times E = -\partial B/\partial t$$

Gives the curl of E at every point in space due to time-varying B. Valid at every point, not just along a path.

Shortcoming of Differential Form:

- Requires knowledge of field vectors at every point — harder to apply practically
- Cannot be used directly to calculate induced voltage in a circuit
- Applies only at a mathematical point — does not directly give macroscopic EMF

Physical Meaning:

A time-varying magnetic field creates a non-conservative electric field. The field lines of E form closed loops (unlike electrostatic E which originates from charges).

Q15. Difference Between Conduction and Displacement Current via Maxwell's Equations

From Maxwell's Equation:

$$\nabla \times H = J + J_d \quad \text{where } J_d = \partial D/\partial t$$

Both J and Jd produce the same effect — they create a magnetic field. But their nature differs:

Conduction Current ($J = \sigma E$):

- Due to free electrons moving through a conductor under E
- Exists only in materials with free charges (conductors, semiconductors)
- Produces ohmic heating — power density $P_v = J \cdot E = \sigma E^2 = J^2/\sigma$ (W/m³)
- In phase with E field
- Associated with DC and AC circuits

Displacement Current ($J_d = \partial D/\partial t = \varepsilon \partial E/\partial t$):

- Not actual charge movement — a fictitious current
- Exists wherever E field is time-varying — including vacuum
- No ohmic heating associated with it
- Leads E by 90° in time (since it's the time derivative of D)
- Responsible for EM wave propagation in free space

Maxwell's Contribution:

Maxwell added Jd to Ampere's law to make it consistent with the continuity equation and to predict that changing E fields in a capacitor (where no conduction current flows) still produce magnetic fields — completing the loop for electromagnetic wave propagation.

Q16. Numerical: Four 40μC Charges at Square Corners — Force on 200μC at Centre +5m

Given:

- Four charges $q = 40\ \mu C = 40\times10^{-6}$ C at corners of a square
- Square diagonal d = 12 m → side $a = d/\sqrt{2} = 12/\sqrt{2} = 8.485$ m
- Half-diagonal = 6 m (distance from center to each corner)
- Test charge $Q = 200\ \mu C = 200\times10^{-6}$ C, located 5 m above center

Distance from each corner charge to Q:

$$r = \sqrt{(6)^2 + (5)^2} = \sqrt{36 + 25} = \sqrt{61} \approx 7.81 \text{ m}$$

Force from one corner charge on Q:

$$F_1 = kqQ/r^2 = (9\times10^9 \times 40\times10^{-6} \times 200\times10^{-6}) / 61$$

$$F_1 = (9\times10^9 \times 8\times10^{-9}) / 61 = 72/61 \approx 1.180 \text{ N}$$

Direction of each force:

Each force F_1 is directed from the corner charge to Q. The horizontal (x-y) components cancel by symmetry (4 symmetric charges). Only the vertical (z) components add.

Vertical component of each force:

$$\cos\theta = 5/7.81 = 0.6402$$

$$F_1z = F_1 \times \cos\theta = 1.180 \times 0.6402 \approx 0.755 \text{ N}$$

Total Force (all 4 charges):

$$F\_total = 4 \times F_1z = 4 \times 0.755 \approx 3.02 \text{ N} \quad \text{(upward, in +z direction)}$$

The net force on the 200 μC charge is approximately 3.02 N directed upward (away from the plane of charges).

Q17. In Free Space, V = 6xy²z + 8. At P(1,2,-5) Find E and Volume Charge Density

Step 1: Find Electric Field E = -∇V

$$E = -(\partial V/\partial x\ \hat{a}_x + \partial V/\partial y\ \hat{a}_y + \partial V/\partial z\ \hat{a}_z)$$
$$\partial V/\partial x = 6y^2z$$
$$\partial V/\partial y = 12xyz$$
$$\partial V/\partial z = 6xy^2$$
$$E = -6y^2z\ \hat{a}_x - 12xyz\ \hat{a}_y - 6xy^2\ \hat{a}_z \quad V/m$$

At P(1, 2, -5):

$$E = -6(4)(-5)\ \hat{a}_x - 12(1)(2)(-5)\ \hat{a}_y - 6(1)(4)\ \hat{a}_z$$
$$E = 120\ \hat{a}_x + 120\ \hat{a}_y - 24\ \hat{a}_z \quad V/m$$
$$|E| = \sqrt{(120^2 + 120^2 + 24^2)} = \sqrt{(14400+14400+576)} = \sqrt{29376} \approx 171.4\ V/m$$

Step 2: Find Volume Charge Density ρv

Using Poisson's equation: $\nabla^2 V = -\rho v/\varepsilon_0$

$$\nabla^2 V = \partial^2V/\partial x^2 + \partial^2V/\partial y^2 + \partial^2V/\partial z^2$$
$$\partial^2V/\partial x^2 = 0 \quad (\partial/\partial x \text{ of } 6y^2z = 0)$$
$$\partial^2V/\partial y^2 = 12xz$$
$$\partial^2V/\partial z^2 = 0$$
$$\nabla^2 V = 12xz$$

At P(1, 2, -5): $\nabla^2 V = 12(1)(-5) = -60$

$$\rho v = -\varepsilon_0\ \nabla^2 V = -8.854\times10^{-12} \times (-60) = 531.2\ pC/m^3$$

Q18. Characteristics of Equipotential Surfaces

An equipotential surface is a surface on which the electric potential V is the same everywhere.

Characteristics:

- No work is done in moving a charge along an equipotential surface — since ΔV = 0, W = qΔV = 0.
- Electric field lines are always perpendicular to equipotential surfaces at every point.
- Equipotential surfaces never intersect each other (if they did, a point would have two potentials simultaneously — impossible).
- The spacing between equipotential surfaces indicates the strength of E: closer spacing → stronger field.
- For a point charge, equipotential surfaces are concentric spheres.
- For a uniform field, equipotential surfaces are parallel planes.
- Conductors in electrostatic equilibrium are equipotential bodies — their entire surface (and interior) is at the same potential.
- E is always in the direction of decreasing potential: E = -∇V

Q19. What are Gaussian Surfaces?

A Gaussian surface is an imaginary, closed mathematical surface chosen to apply Gauss's Law conveniently in order to calculate the electric field or flux.

Properties:

- It is always a closed surface (encloses a volume)
- It must be chosen such that E is either constant and parallel to dS, or perpendicular (zero flux) on various portions
- It is a mathematical tool — not a physical surface

Common Gaussian Surfaces:

- **Sphere:** For point charges or spherically symmetric charge distributions
- **Cylinder:** For infinite line charges or cylindrical charge distributions
- **Rectangular box (pillbox):** For infinite planar charge distributions

Application:

For a charge distribution with sufficient symmetry, Gauss's Law gives:

$$\oint D \cdot dS = Q_{enc} \quad \rightarrow \quad D \times A_{surface} = Q_{enc}$$

This allows direct calculation of D (and E) without integration along complex paths.

Q20. Concept of Work Done — Explain Potential

Work Done in Electrostatics:

Work done in moving a test charge Q from point A to point B in an electric field E is:

$$W = -Q \int_{(A \to B)} E \cdot dL \quad \text{(Joules)}$$

The negative sign indicates that work is done against the electric force when moving from lower to higher potential.

Electric Potential (V):

Electric potential at a point is defined as the work done per unit positive charge in moving it from a reference point (usually infinity) to that point, against the electric field:

$$V = W/Q = -\int_{(\infty \to P)} E \cdot dL \quad \text{(Volts)}$$

Potential Difference:

$$V_{AB} = V_A - V_B = -\int_{(B \to A)} E \cdot dL = \int_{(A \to B)} E \cdot dL$$

VAB represents the work done per unit positive charge in moving from B to A. It is path-independent in a static (conservative) field.

Key Points:

- Potential is a scalar quantity — easier to work with than vector E
- E and V are related by: $E = -\nabla V$
- For a point charge: $V = Q / (4\pi\varepsilon r)$
- The reference point ($V = 0$) is usually taken at infinity for isolated charges
- Work done in a closed path (conservative field) = 0: $\oint E \cdot dL = 0$

Q21. Explain the Concept of Gradient of a Potential Function

The gradient of a scalar function gives a vector that points in the direction of maximum rate of increase of that function, with magnitude equal to that maximum rate.

Mathematical Definition:

For a scalar potential V(x, y, z), the gradient in Cartesian coordinates is:

$$\nabla V = \partial V/\partial x \; \hat{a}_x + \partial V/\partial y \; \hat{a}_y + \partial V/\partial z \; \hat{a}_z$$

Relationship with Electric Field:

The electric field E is the negative gradient of the potential:

$$E = -\nabla V$$

The negative sign indicates that E points in the direction of maximum decrease of V (from high to low potential).

Physical Interpretation:

- ∇V is perpendicular to equipotential surfaces
- The magnitude $|\nabla V|$ gives the rate of change of potential with distance
- If V is constant in a direction, the E component in that direction is zero

- The gradient converts a scalar field (V) into a vector field (E)

In Cylindrical and Spherical Coordinates:

$$\text{Cylindrical: } \nabla V = \partial V/\partial r\ \hat{a}r + (1/r)\partial V/\partial \varphi\ \hat{a}\varphi + \partial V/\partial z\ \hat{a}z$$
$$\text{Spherical: } \nabla V = \partial V/\partial r\ \hat{a}r + (1/r)\partial V/\partial \theta\ \hat{a}\theta + (1/r\ \sin\theta)\partial V/\partial \varphi\ \hat{a}\varphi$$

Q22. Explain Conservative Field Property — Which Maxwell's Equation Explains It?

Conservative Field:

A vector field F is called conservative if the line integral of F around any closed path is zero:

$$\oint F \cdot dL = 0$$

This means the work done in moving between two points is path-independent — it depends only on the start and end points.

Electrostatic Field is Conservative:

The static electric field E is conservative because no matter what path is taken from A to B:

$$\int(A\to B)\ E \cdot dL = VA - VB \quad \text{(path independent)}$$
$$\oint E \cdot dL = 0 \quad \text{(for static fields)}$$

Maxwell's Equation That Explains This:

Faraday's Law in differential form:

$$\nabla \times E = -\partial B/\partial t$$

For static fields (no time variation), $\partial B/\partial t = 0$, so:

$$\nabla \times E = 0$$

A field with zero curl is irrotational, and an irrotational field is conservative. This means a static electric field can be expressed as the gradient of a scalar potential: $E = -\nabla V$.

Physical Implication:

- The work done in moving a charge in a static E field around a closed loop is always zero
- Energy is conserved in static EM systems
- A potential function V can always be defined for static E fields
- In time-varying fields, $\nabla \times E \neq 0$, so E is not conservative — this gives rise to EMF in transformers and generators

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End of MSE Review Answers — Good luck on your exam!